ACHARYA INSTITUTE OF TECHNOLOGY

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Department of Computer Science & Engineering (Data Science)

PROJECT REPORT
ON
Flipkart: Big Data Analytics for "The Big Billion Days"

Subject Name: BIG DATA ANALYTICS
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1. Task 1: Big Data in Daily Life – Visual Storytelling

Infographic: The 5 V's of Flipkart's Big Billion Days

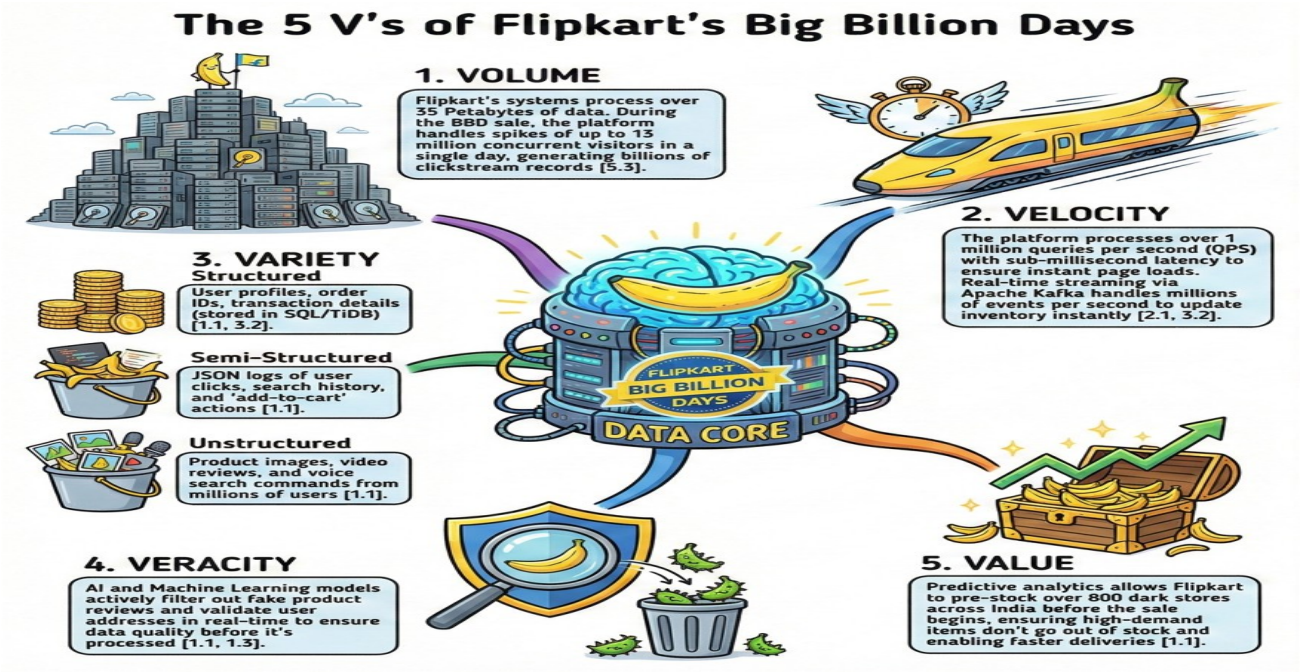

Figure 1.1:Infographic of Flipkart Big Data Characteristics

Data Classification:
Structured: Transaction details, Order IDs (SQL).
Semi-Structured: Clickstream logs, Search history (JSON).
Unstructured: Product images, User review videos (HDFS).

The 5 V's at Flipkart:
Volume: 35+ Petabytes of data storage.
Velocity: 1 Million events per second during flash sales.
Variety: Handling text, images, and streaming video.
Veracity: ML-driven fraud and fake review detection.
Value: Real-time demand forecasting for 24-hour delivery.

Comic Strip: The Flash Sale Hero

Figure 1.2:Comic strip illustrating the transition from RDBMS to Big Data during Flipkart's BBD

2. Task 2: BI vs. Big Data – A Strategic Dialogue

Setting: A meeting room at Flipkart headquarters a few weeks before the Big Billion Days sale. The Manager and Consultant are discussing whether the company really needs to move to a Big Data system.

Manager: Honestly, I don't understand why we are shifting everything to this "Big Data" system. I've been using Excel for years and it works perfectly fine for reports.

Consultant: Excel works for small businesses, but we are talking about Flipkart during Big Billion Days. The traffic is massive and keeps increasing every year.

Manager: But we already have SQL databases. We just run queries and generate sales reports. Why complicate things?

Consultant: SQL works well for structured data, but during flash sales, millions of users are clicking at the same time. Traditional databases struggle with that load.

Manager: Can't we just upgrade the server? Add more RAM and better processors?

Consultant: That's called vertical scaling. It helps, but only up to a limit. You can't keep upgrading one single machine forever.

Manager: So what does Big Data do differently?

Consultant: Big Data uses horizontal scaling. Instead of one powerful machine, it uses many smaller machines working together as a cluster.

Manager: And that's where Hadoop comes in, right?

Consultant: Exactly. Hadoop helps store and process huge amounts of data across multiple machines. It can handle structured, semi-structured, and unstructured data.

Manager: What do you mean by unstructured data?

Consultant: Things like product images, customer reviews, voice searches, videos, and even live location tracking. This type of data doesn't fit neatly into tables.

Manager: But I only need daily sales reports. Why do we need real-time data?

Consultant: During flash sales, products sell out in seconds. If we wait till tomorrow, we lose the chance to restock or promote alternatives instantly.

Manager: That makes sense. So this is about faster decision-making?

Consultant: Yes, and smarter decision-making. With streaming analytics, we can see what customers are viewing at that exact moment.

Manager: Is that how I get those "Recommended for You" notifications?

Consultant: Yes! The system analyzes your browsing behavior and suggests products instantly to increase sales.

Manager: Interesting. And what about NoSQL? I keep hearing that term.

Consultant: SQL databases require fixed tables and structure. NoSQL is flexible. If tomorrow we add video reviews or new customer details, NoSQL can handle it easily without changing the entire system.

Manager: But isn't storing data on many machines risky? What if one crashes?

Consultant: That's the best part. Hadoop is fault-tolerant. It automatically stores copies of data on different machines. If one fails, the system keeps running without data loss.

Manager: Okay, now I see why Excel and traditional systems are not enough for such a large platform.

Consultant: Exactly. Big Data helps us move from "What happened yesterday?" to "What is happening right now?" and even "What will happen next?" That's the real advantage.

3. Task 3 – Architecture Design Challenge

Traditional Data Warehouse Architecture

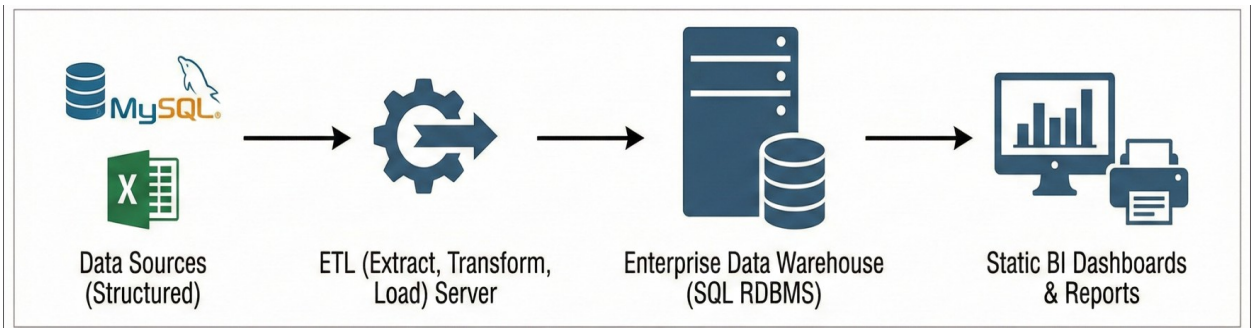

Figure 3.1:Traditional Data Warehouse Architecture for Retail

In the traditional retail model, Flipkart would rely on a centralized relational system. This architecture is designed for "Small Data" and structured transactions only. It uses a "Schema-on-Write" approach, meaning data must be perfectly formatted before it is saved.

Data Sources: Transactional databases (MySQL), ERP systems, and manual CSV files from sellers.

Storage: A central Relational Database Management System (RDBMS) like SQL Server or Oracle.

Processing Layer: Batch-oriented ETL (Extract, Transform, Load) processes that usually run during off-peak hours (e.g., midnight).

Analytics Layer: Static Business Intelligence tools used for descriptive reporting (e.g., "What were the total sales yesterday?").

Hadoop-Based Big Data Architecture (The Flipkart BBD Solution)

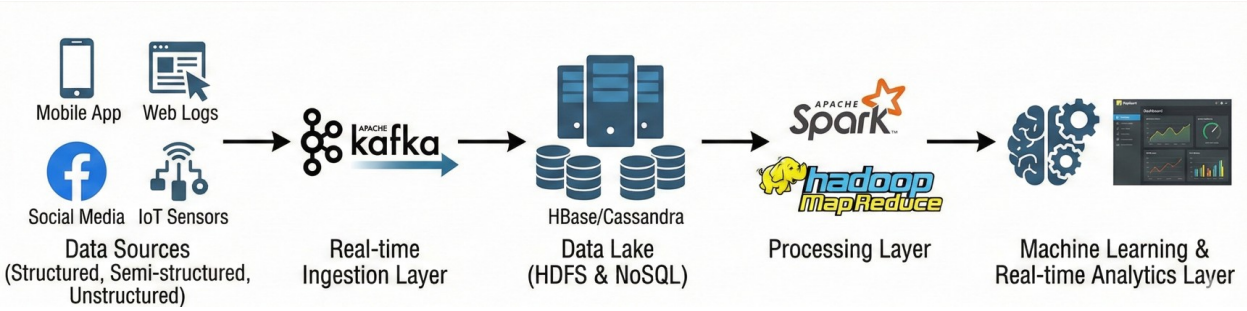

Figure 3.2:Modern Hadoop-Based Architecture for High-Velocity E-commerce

To handle the "Big Billion Days," a distributed architecture is used. This follows the "Schema-on-Read" principle, allowing for massive variety and velocity.

Data Sources: Clickstream logs (User clicks), Mobile App events, IoT sensors from warehouses, and Social Media feeds.

Ingestion (Buffer): Apache Kafka – This layer acts as a shock absorber, catching millions of events per second so the database doesn't crash.

Storage: HDFS (Hadoop Distributed File System)-for low-cost, fault-tolerant storage of raw data.

Processing Layer: Apache Spark – Used for real-time processing and complex transformations in-memory, which is 100x faster than MapReduce.

Analytics Layer: Predictive AI models, real-time dashboards for monitoring flash sales, and personalized recommendation engines.

4. Task 4: Analytics & Tool Match

To effectively manage the scale of "The Big Billion Days," Flipkart utilizes a combination of distributed storage, real-time processing engines, and machine learning libraries. The table below maps specific business questions to their respective analytics types and the Big Data tools required to execute them.

Business Question	Analytics Type	Big Data Tools	Flipkart Context
What happened?	Descriptive	Hadoop (HDFS) + Hive	Analyzing total sales volume after the first hour of a flash sale.
Why did it happen?	Diagnostic	Apache Spark	Identifying that a specific bank offer caused a 40% surge in smartphone checkouts.
What will happen next?	Predictive	Spark MLlib / Python	Forecasting website traffic spikes at midnight to allocate server resources.
What action should be taken?	Prescriptive	Apache Flink / Kafka	Automatically triggering a "Low Stock" alert to reroute inventory to a high-demand hub.

Table 1: Mapping Business Logic to Big Data Tools

Justification of Tool Selection:

- **Hadoop/Hive:** Used for batch processing of historical data to understand long-term trends.
- **Apache Spark:** Chosen for its in-memory processing capabilities, making it 100 times faster than traditional MapReduce for diagnostic tasks.
- **NoSQL (HBase):** Utilized when low latency access to user profiles and "Wishlist" data is required during high traffic periods.
- **Machine Learning (MLlib):** Essential for moving from simple reporting to predictive modeling, ensuring Flipkart stays ahead of customer demand.

5. Bonus Challenge: Explain Big Data to a 10-Year-Old

The World's Busiest Post Office

Imagine you have a tiny neighborhood post office with just one mailman. If ten people drop off a letter, he can easily tuck them into his bag, whistle a tune, and deliver them all before lunchtime. He knows everyone's name and exactly where they live. This is Normal Data. But now, imagine it is the morning of the biggest holiday of the year, and suddenly 10 million people drop a letter into the box at the exact same second! If that one mailman tries to pick them all up, he will be buried under a mountain of paper as high as a skyscraper. He would get totally overwhelmed, lose the addresses, and the post office would have to lock its doors. This is exactly what happens to a regular computer during a giant Flipkart sale; it just isn't strong enough to carry that heavy mountain of orders all by itself.

Big Data is like turning that tiny room into a "Super Post Office" with thousands of mailmen all working together as one giant team. Instead of one person struggling alone, the work is shared perfectly. One group of mailmen uses super-fast scanners to sort the letters by color, while another group uses a "Magic Satellite Map" to find secret shortcuts across the city so they never get stuck in traffic. They even have smart helpers who can guess which houses will send letters next so they can be ready and waiting! Because they all work at the exact same time, every single letter reaches the right front door perfectly, even though there were millions of them flying around at once. That is how Flipkart works. Big Data is not just a giant pile of mail; it is a massive army of smart computers working as a team so that when you click "Buy" on a new gadget, it never gets lost in the crowd.